Pharmacotherapy 2: Ultimate Midterm Bank

Part 1: High-Yield Extracted Questions (Matches PDF Format)

1. Which chemical mediators are released upon tissue damage to directly stimulate pain receptors?

A) Prostaglandins and Leukotrienes

B) Histamine and Bradykinin

C) Acetylcholine and Dopamine

D) Serotonin and GABA

> 💡 **Hint:** As the doctor explained with the 'nail' example, tissue injury causes immediate release of Histamine and Bradykinin which trigger pain. Prostaglandins only 'sensitize' the receptors, they don't stimulate them directly.

2. Why can NSAIDs precipitate severe bronchospasm in asthmatic patients?

A) They inhibit Leukotriene synthesis

B) They shunt Arachidonic Acid to the LOX pathway, massively increasing Leukotrienes

C) They increase Histamine release from mast cells

D) They block Beta-2 receptors in the lungs

> 💡 **Hint:** This is the 'Asthma Trap'. Blocking the COX pathway forces all Arachidonic Acid into the Lipoxygenase (LOX) pathway, producing excess Leukotrienes which are potent bronchoconstrictors.

3. What makes Aspirin unique among other NSAIDs regarding its mechanism of action on platelets?

A) It selectively inhibits COX-2 only

B) It irreversibly inhibits COX-1 and COX-2 via covalent acetylation

C) It competitively and reversibly blocks COX-1

D) It acts centrally without affecting peripheral tissues

> 💡 **Hint:** Aspirin is 'irreversible'. It permanently alters the COX enzyme in platelets for the platelet's entire lifetime (7-10 days), which is why it must be stopped a week before surgery.

4. The risk of Reye's syndrome occurs if which of the following is given to children under the age of 15 with a viral infection?

A) Diflunisal

B) Ibuprofen

C) Aspirin

D) Acetaminophen

> 💡 **Hint:** Aspirin is strictly contraindicated in children with viral infections (like flu or chickenpox) due to the risk of fatal Reye's syndrome (liver and brain damage). Acetaminophen is the safe alternative.

5. Paracetamol (Acetaminophen) differs from true NSAIDs primarily because it has:

A) Strong peripheral anti-inflammatory activity

B) Irreversible COX-1 inhibition

C) Significant GI toxicity and ulceration risk

D) Weak or no peripheral anti-inflammatory effect

> 💡 **Hint:** Acetaminophen works centrally in the brain to reduce pain and fever, but is inactivated by peroxides in inflamed peripheral tissues, making it a very weak anti-inflammatory drug.

6. In cases of severe Acetaminophen overdose, liver failure occurs due to the depletion of Glutathione. What is the specific antidote?

A) Atropine

B) N-acetylcysteine (NAC)

C) Flumazenil

D) Sodium Bicarbonate

> 💡 **Hint:** Overdose produces the toxic metabolite NAPQI. N-acetylcysteine provides the sulfhydryl groups needed to replenish glutathione and neutralize the toxin. Must be given within 10 hours.

7. Patients with normal body temperature who take Aspirin will experience:

A) Severe hypothermia

B) A slight decrease in body temperature

C) No effect on normal body temperature

D) Rebound hyperthermia

> 💡 **Hint:** NSAIDs reset the elevated hypothalamic 'thermostat' back to normal during a fever, but they do NOT lower a normal body temperature.

8. Which of the following is considered a first-line treatment for an Acute Gout Attack?

A) Allopurinol

B) Probenecid

C) Indomethacin or Colchicine

D) Methotrexate

> 💡 **Hint:** Acute attacks require anti-inflammatory action. Indomethacin (NSAID) or Colchicine (inhibits neutrophil migration) are used. Allopurinol is for chronic prophylaxis, NOT acute attacks.

9. Why is Aspirin contraindicated during an Acute Gout attack?

A) It interacts fatally with Colchicine

B) Low doses compete with uric acid for renal secretion, worsening hyperuricemia

C) It heavily increases uric acid synthesis in the liver

D) It causes rapid crystal dissolution leading to kidney stones

> 💡 **Hint:** Aspirin blocks the organic acid secretion mechanism in the proximal tubule, which reduces uric acid excretion and makes the gout attack worse.

10. Allopurinol is the drug of choice for chronic gout prophylaxis in 'overproducers'. What is its mechanism?

A) It blocks proximal tubular reabsorption of uric acid

B) It inhibits Xanthine Oxidase, reducing uric acid synthesis

C) It binds to tubulin, preventing leukocyte migration

D) It dissolves sodium urate crystals directly

> 💡 **Hint:** Allopurinol is a purine analog that competitively inhibits Xanthine Oxidase, the enzyme responsible for the final steps of uric acid biosynthesis.

11. Triptans (like Sumatriptan) are used for acute migraine attacks. They work by acting as agonists at which specific receptors?

A) 5-HT2A

B) 5-HT1B and 5-HT1D

C) Dopamine D2

D) Alpha-1 adrenergic

> 💡 **Hint:** Triptans are 5-HT1D/1B agonists that cause rapid vasoconstriction of painfully dilated cranial blood vessels. They are strictly contraindicated in Ischemic Heart Disease.

12. The primary cause of death in acute Opioid overdose is:

A) Severe constipation and bowel obstruction

B) Fatal arrhythmias

C) Respiratory Depression (decreased sensitivity to CO_2)

D) Hypertensive crisis

> 💡 **Hint:** Opioids depress the respiratory center in the brainstem, making it insensitive to Carbon Dioxide (CO_2). The patient simply 'forgets' to breathe.

13. Which of the following is a diagnostic clinical sign of Morphine overdose?

A) Mydriasis (Dilated pupils)

B) Miosis (Pinpoint pupils)

C) Tachycardia

D) Hyperventilation

> 💡 **Hint:** Miosis (pinpoint pupils) is a cardinal sign of opioid overdose. The doctor emphasized that tolerance NEVER develops to miosis and constipation.

14. Unlike Morphine, Meperidine (Pethidine) administration can cause which of the following eye effects due to its unique receptor activity?

A) Severe Miosis

B) Mydriasis (Pupil dilation)

C) Nystagmus

D) Cataracts

> 💡 **Hint:** Meperidine has 'atropine-like' (anticholinergic) properties, so it dilates the pupil (Mydriasis), contrasting sharply with morphine's pinpoint pupils.

15. A patient with a known history of severe Bronchial Asthma requires a strong analgesic. Why should Morphine be used with extreme caution or avoided?

A) It blocks Beta-2 receptors

B) It causes severe Histamine release leading to bronchoconstriction

C) It directly stimulates Leukotriene synthesis

D) It induces severe coughing fits

> 💡 **Hint:** Morphine triggers the release of histamine from mast cells, which can provoke a fatal bronchospasm in asthmatic patients.

16. Which drug is an IV pure opioid antagonist used specifically for the emergency reversal of an opioid overdose coma?

A) Methadone

B) Naltrexone

C) Naloxone

D) Buprenorphine

> 💡 **Hint:** Naloxone is the rapid-acting IV rescue drug. Naltrexone is also an antagonist but is long-acting and given orally for maintenance/relapse prevention.

17. Typical (First-Generation) antipsychotics like Haloperidol primarily exert their therapeutic effect by blocking:

A) Serotonin 5-HT2A receptors

B) Histamine H1 receptors

C) Dopamine D2 receptors in the mesolimbic system

D) Muscarinic M1 receptors

> 💡 **Hint:** They are competitive D2 receptor blockers. This effectively reduces positive symptoms but drastically increases the risk of Extrapyramidal Symptoms (EPS).

18. A patient on Haloperidol develops severe muscle rigidity, high fever, altered mental status, and unstable blood pressure. What is the most likely diagnosis?

A) Tardive Dyskinesia

B) Neuroleptic Malignant Syndrome (NMS)

C) Serotonin Syndrome

D) Acute Dystonia

> 💡 **Hint:** NMS is a rare but fatal idiosyncratic reaction to neuroleptics. Treatment requires immediate discontinuation and administration of Dantrolene or Bromocriptine.

19. Why do Atypical antipsychotics (like Risperidone and Clozapine) have a much lower risk of causing Extrapyramidal Symptoms (EPS)?

A) They do not block Dopamine receptors at all

B) They completely destroy the Nigrostriatal pathway

C) They strongly block Serotonin (5-HT2A) receptors alongside weak D2 blockade

D) They are rapidly excreted by the kidneys

> 💡 **Hint:** The dual action (blocking 5-HT2A and D2/D4) is the hallmark of Atypicals. This profile treats negative symptoms better and heavily minimizes EPS.

20. Which atypical antipsychotic is reserved only for refractory Schizophrenia due to its 1-2% risk of causing potentially fatal Agranulocytosis?

A) Olanzapine

B) Ziprasidone

C) Clozapine

D) Quetiapine

> 💡 **Hint:** Clozapine requires extremely strict regular blood monitoring (CBC) because it can severely drop White Blood Cell counts (Agranulocytosis).

21. Which Anti-Epileptic Drug (AED) is considered 'Broad-Spectrum' and is highly effective across all seizure types, including Absence seizures?

A) Carbamazepine

B) Phenytoin

C) Valproic Acid

D) Ethosuximide

> 💡 **Hint:** Valproic Acid works via multiple mechanisms (Na+ block, T-type Ca++ block, GABA increase), making it the ultimate broad-spectrum choice. (Carbamazepine actually worsens Absence seizures).

22. A pregnant woman taking Valproic Acid is at a significantly higher risk of having a child with which specific teratogenic defect?

A) Cleft lip

B) Spina Bifida (Neural tube defects)

C) Congenital heart block

D) Phocomelia

> 💡 **Hint:** Valproic acid is heavily teratogenic and causes Spina Bifida. Pregnant women must be managed with alternative safer AEDs or high-dose folic acid if unavoidable.

23. Phenytoin exhibits 'Zero-Order Kinetics' at high therapeutic doses. What does this mean clinically?

A) Its half-life decreases as the dose increases

B) It is not metabolized by the liver at all

C) Very small dose increments can cause massive, unpredictable increases in plasma levels and toxicity

D) It completely stops acting after 24 hours

> 💡 **Hint:** Zero-order kinetics means the liver enzymes are fully saturated. Adding even a tiny extra dose cannot be processed, causing blood levels to spike dangerously.

24. A young child on chronic Phenytoin therapy is most likely to develop which of the following cosmetic/dental side effects?

A) Severe dental caries

B) Gingival hyperplasia (Gum overgrowth)

C) Blue discoloration of teeth

D) Loss of tongue papillae

> 💡 **Hint:** Gingival hyperplasia is a classic and very common side effect of Phenytoin, especially in pediatric patients. Good oral hygiene is required.

25. What is the most dangerous consequence of abruptly stopping Anti-Epileptic Drug (AED) therapy?

A) Permanent memory loss

B) Severe hypotension

C) Precipitation of Status Epilepticus

D) Drug-induced Parkinsonism

> 💡 **Hint:** The doctor strongly emphasized the 'Don't Stop' rule. Abrupt withdrawal removes the seizure threshold protection instantly, throwing the brain into continuous, life-threatening seizures (Status Epilepticus).